# INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

## CHAPTER 1: Introduction and Research Problem

by

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# 1 Introduction and Research Problem

A public blockchain that runs smart contracts holds very many of them. On Arbitrum One, the `Approval` and `Transfer` events of five tokens between April 2025 and September 2026 alone involve 1,829,719 addresses that this study types as contracts (Table 8.4). Wallet applications, block explorers, auditors and integrators cannot examine them all and must choose which contracts to look at first.

This thesis asks whether a PageRank walk over the relations between contracts finds the contracts that users rely on, as PageRank ranked the pages of the web by the links they receive (Page et al., 1999). The relations are ERC-20 transfers from one contract to another and allowances, by which an owner contract authorizes a spender contract to move its tokens. The method of the thesis, a coupled PageRank (C-PR), walks both edge types in one chain and mixes them at each contract with a fixed weight $\lambda$. Two single-layer walks serve as its references: EndorseRank, PageRank on the allowance edges alone with the edge weights of the transfer-based Adaptive Weighted PageRank (AWP) of T. Do et al. (2023), and AWP itself on the transfers. Every node of every graph is a contract. The walks are evaluated beside the raw counts that they smooth: by how soon a crawl ordered by each score reaches the contracts that receive the most transfers and approvals from user accounts, by the approvals and inflow that arrive after the scores are frozen, and under the decision rules of an analysis plan and a registration.

## 1.1 Background: contracts on a rollup and how trust in them is formed

Public blockchains make programmable settlement possible without a central authority (Nakamoto, 2008), and protocols stack smart contracts (Szabo, 1997) into decentralized markets for lending, automated market making and perpetual futures (Werner et al., 2022), composing new financial

services from existing ones (Kitzler et al., 2023). On Arbitrum One and other Layer-2 (L2) networks fees are low, and every transfer, approval and protocol event can be inspected address by address.

Whoever deals with these contracts has to decide which of them deserve attention, and today that decision rests on evidence about one contract at a time. A wallet application asks its user to confirm an approval for a spender that the user may never have seen, and phishing sites lead users to sign transactions that let scammers take their assets (He et al., 2023). Code analysis looks in a contract's code for malicious functions (Z. Lin et al., 2024), an audit examines one protocol, and lists of known contracts are curated by hand. None of these reads which contracts rely on which, although the logs make that structure public.

The web met the same problem at a larger scale. PageRank ranks a page by the pages that link to it and weights each link by the rank of its source (Page et al., 1999). Cho et al. (1998) put such rankings to a test of discovery: a crawler knows only the links of the pages it has read and must choose which known page to read next. They compared orderings of that choice, breadth-first order, the number of backlinks and PageRank, the last two computed on the pages seen so far, and on a stored copy of the Stanford University web the crawl ordered by PageRank reached the hot pages, those with many backlinks, sooner than the crawl ordered by the backlink count (Section 2.6).

On public ledgers, PageRank has been applied to transfers: H.-D. Do and Do (2023) run PageRank and HodgeRank on Ethereum transactions, and AWP (T. Do et al., 2023) weights each transfer by its value and age, restarts the walk at addresses that sent recently, and is validated by its correlation with inbound transfer count and value. These walks run over the transfers among all addresses, user accounts included, and user accounts supply much of what a contract receives: more than four in five of the new approval pairs and new transfer senders that the spender contracts of this study gain in its label windows come from user accounts (Table 8.6). Anyone can create such accounts at almost no cost, and the study does not measure who controls them. A contract costs a deployment, and an edge between two contracts records a relation that one program holds toward another. An ERC-20 allowance between two contracts is such a relation, a standing authorization under which

the spender can move the owner's tokens (Vogelsteller & Buterin, 2015), and the transfer walks do not read it. Figure 1.1 sets the web, these transfer walks and the graphs of this thesis side by side.

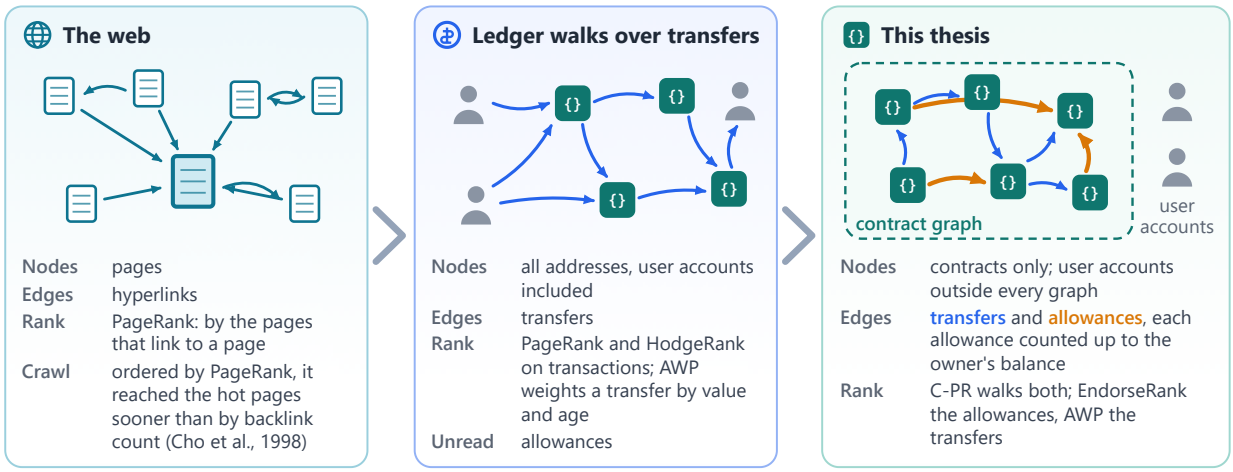



Figure 1.1: From links between web pages to records between contracts. On the web, PageRank ranks a page by the pages that link to it, and in the crawl-ordering experiment of Cho et al. (1998) a crawl ordered by PageRank on the pages read so far reached the hot pages sooner than a crawl ordered by the backlink count. Ledger walks such as AWP (T. Do et al., 2023) run over the transfers among all addresses, user accounts included, and do not read allowances. The graphs of this thesis hold contracts only, with transfers (blue) and allowances (orange) between them, each allowance counted up to its owner's balance; user accounts stand outside every graph.

## 1.2 Motivation for on-chain reputation

Three considerations motivate a ranking of contracts that is computed from public logs.

- Attention: wallet applications, explorers, auditors and integrators need an order in which to examine contracts. A ranking that puts first the contracts that users rely on tells them where to start; it replaces no examination (Section 5.4).
- Auditability: anyone can recompute a score that a fixed rule derives from public event data, unlike automated scores that outsiders cannot inspect (Packin & Lev-Aretz, 2024).
- Cost: a score that is refreshed often should be cheap to recompute, and one that user accounts can move is cheap to manipulate. In a graph of contracts every node costs a deployment, and an allowance counted only up to its owner's balance has to be backed by tokens (Section 3.6).

A ranking that proved reliable in these respects could later serve as one input to credit decisions about contracts; the thesis does not test that use.

Transfer-based AWP is public and auditable, but its graph grows with every transfer (T. Do et al., 2023). An allowance graph is built differently. For an owner, a spender and a token, each new approval supersedes the older ones (Vogelsteller & Buterin, 2015), so the latest approval stands for the history of that pair. An allowance can exceed the owner's balance or be unlimited, yet the spender can move at most the balance, so each latest allowance is counted only up to its owner's balance of the token at the freeze (the balance cap, Section 3.6). Between contracts the resulting graph $G_{\text{approve}}$ has far fewer edges than the transfer graph $G_{\text{transfer}}$, so its PageRank solve takes less time, a property of the smaller graph and not of the algorithm (Table 4.3; Claim C6). The cap also leaves the layer thin: most latest allowances between contracts belong to owners that hold no balance of the token and form no edge (Table 8.5).

Allowances also mean something different. A transfer might reflect routing, market making, or a passive receipt, whereas an allowance is an active grant of permission to spend funds, so treating allowances as endorsement edges carries PageRank's citation reading (Page et al., 1999) over to a trust relationship that protocols already depend on. Whether one walk over both kinds of edge finds the contracts that users rely on is examined on the contract graph of Arbitrum One and on a matched cohort of $n = 6{,}443$ of its contracts.[1]

## 1.3 Research problem

The work reviewed offers those who must choose which contracts to examine no ranking computed from the relations among contracts alone, and it is not known whether such a ranking would find the contracts that users rely on. AWP infers trust from the transfer history of all addresses and assigns no role to spending authorization (T. Do et al., 2023). EndorseRank, the allowance reference of this thesis, keeps AWP's edge weights and solver but builds the directed graph from the latest allowance

[1] Every method scores the same contracts: those that received a non-zero approval of WETH, USDC, USDT0, ARB or WBTC from at least three distinct owners, of any account type, between 1 July 2025 and 30 June 2026 (Section 3.2).

of each owner–spender pair and token. An allowance points from the owner to the spender, so EndorseRank scores the spenders that receive approvals and gives every owner that receives none the same value. Each of the two walks therefore follows one kind of act: AWP the transfers that reach a contract, EndorseRank the spending rights it is granted.

Because both edge types live on the same contracts, a single walk can follow both, mixing them node by node with a fixed weight $\lambda$. This walk, the Coupled PageRank (C-PR), is the method of the thesis; at $\lambda = 1$ and $\lambda = 0$ it walks one layer alone. The main research question is: does one PageRank walk over allowance and transfer edges between Arbitrum One contracts find the contracts that users rely on sooner, and predict their later use better, than single-layer walks and raw counts? Research questions RQ1–RQ5 (Section 1.6) break it down. The analysis plan fixed one decision rule, and its object is C-PR (Section 3.8).

The first half of the question has a direct test. In the discovery-order test (Section 3.7) a crawl starts at a seed contract, reads the edges of each contract it visits and chooses the next contract by an ordering computed on what it has read so far, as the crawler of Cho et al. (1998) chose its next page; a prime, as in C-PR′, marks such an ordering. Importance is the demand of user accounts, which no ordering reads: the US dollars that user accounts transferred to a contract ($I_T$, user-account inflow) and the balance-capped US dollars that their allowances let it spend ($I_A$, user-authorized value), each capped per account. The decision compares how soon the crawl ordered by C-PR′, a breadth-first crawl and a crawl ordered by the number of contract backlinks reach the contracts with the most of each importance. The test was defined after the registered results were known and decides no rule of the plan or the registration.

The scores appear in Table 1.1. Table 1.2 lists hypotheses (H1–H3), research questions (RQ1–RQ5), and empirical claims (C1–C6) from Chapter 4.

The main terms are defined as follows:

- *Smart contract:* Self-executing program code that runs on a blockchain and carries out an agreement without a trusted intermediary (Szabo, 1997).

Table 1.1: The scores evaluated in this thesis, each computed on the whole contract graph of a freeze.

| Score | Input graph | Role |
|---|---|---|
| C-PR | Both layers of the contract graph, in US dollars with each allowance counted up to its owner's balance and capped, mixed per node with weight $\lambda$ | The method of this thesis, object of the decision of the discovery-order test (as the ordering C-PR′), of rule A and of the two applied criteria of rule B. Its layers walked alone, C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$), are weighted by amounts, so they are not EndorseRank and AWP. Its ablation S-PR restarts a transfer walk from the allowance walk |
| EndorseRank | Latest ERC-20 allowances between contracts on five tokens, in US dollars up to the owner's balance ($G_{\text{approve}}$), weighted as in AWP | Allowance delegation; the single-layer reference on allowances. Its variant with restarts at approving owners is reported beside it |
| AWP | Transfers of five tokens between contracts in US dollars ($G_{\text{transfer}}$) | Transfer flow; the single-layer reference on transfers, the published transfer-graph method of T. Do et al. (2023) (Section 3.6) |
| Raw counts | $G_{\text{approve}}$, $G_{\text{transfer}}$ and their union, counted per node | In-approve degree and value, transfer in-degree and union in-degree (the number of contract backlinks), the local counts that the walks smooth: proxies in the same window (the first three), baselines out of window and comparators in the discovery-order test |

- *Wallet / Address:* A blockchain account that holds funds and interacts with smart contracts, identified by its address (Buterin, 2014): an externally owned account (EOA), controlled by a private key, or a contract account governed by its code (Antonopoulos & Wood, 2018). No off-chain identity is assumed or recovered.
- *User account and contract:* A user account is an address that sent a transaction, an address without code, an EIP-7702 account (Buterin et al., 2024) or a smart account, that is, a Safe (Safe Ecosystem Foundation, n.d.) or an ERC-4337 account (Buterin et al., 2021); every other address with code is a contract (Section 3.5.1). Every node of every graph is a contract, and user accounts count only in the cohort rule, the labels and the importance measures.
- *On-chain reputation score:* A number computed from on-chain data, without off-chain identity information, that expresses a wallet's relative position for financial activity under pseudonymity (H.-D. Do & Do, 2023). The scores here are ordinal, and their only off-chain

inputs are the five-token list and daily US-dollar prices.

- *PageRank:* A graph ranking algorithm that spreads influence through a directed graph with a damping factor (Page et al., 1999). Adaptive Weighted PageRank (AWP) is PageRank on the transfer graph, with edges weighted by the value and age of each transfer and restarts at recently active senders (T. Do et al., 2023). EndorseRank applies AWP's edge weights to the latest-allowance graph and restarts uniformly (Section 3.6).
- *Coupled PageRank (C-PR):* One random walk over both the allowance and the transfer layer, which at each contract mixes the two rows with weights $\lambda$ and $1-\lambda$ over the layers that contract occupies. Endorsement-seeded PageRank (S-PR) keeps the transfer walk but restarts it from the allowance walk, as TrustRank (Gyöngyi et al., 2004) restarts from a seed set.
- *ERC-20 allowance / Approve / Permit:* Contract methods that record how many of the owner's tokens a spender may spend on the owner's behalf (Vogelsteller & Buterin, 2015); EIP-2612 `permit` makes the same state change as `approve` through a signature (Lundfall, 2020). The graphs use, for each owner–spender pair and token, the latest non-zero allowance up to the owner's balance of the token at the freeze (the balance cap); an allowance whose owner holds no balance is no edge.
- *Domain alignment:* The degree to which a reputation ranking agrees with an observable proxy ranking, measured with Spearman's $\rho$ (Spearman, 1904) and Kendall's $\tau$ (Kendall, 1938) and read as evidence on construct validity (Cronbach & Meehl, 1955); with proxies built from a score's own edges it is only an intended-construct check.

## 1.4 Temporal holdout

The contemporaneous allowance Kendall $\tau$ (Table 4.5) is an intended-construct check: PageRank on $G_{\text{approve}}$ should agree with spender-side approval degree, because it smooths that degree globally. To test whether the score conveys information beyond the same-window graph, the scores are frozen at $t_1$ (31 March 2026, 23:59:59 UTC) and compared with two labels that only become available in

April–June 2026: new owner–spender approval pairs (the endorsement construct) and new transfer senders (the flow construct). In this first window, W0, the holdout cohort ($n = 7{,}864$) consists of every contract spender that holds, at $t_1$, at least one positive latest allowance of one of the five tokens set in the twelve months before, among the spenders listed for that freeze (Section 3.8). Each walk is also compared with the raw count it smooths at $t_1$. The labels count every account, most of them user accounts, which no score reads (Table 8.6).

A second window, W1, was registered after the W0 results were known and before the study extracted any log dated after 30 June 2026. Its scores are frozen on 30 June 2026 and its labels counted from July to September 2026, on the $n = 7{,}915$ contract spenders of that freeze (Section 3.8.1). Figure 1.2 places the two freezes, their score windows and the two label windows on one time line.

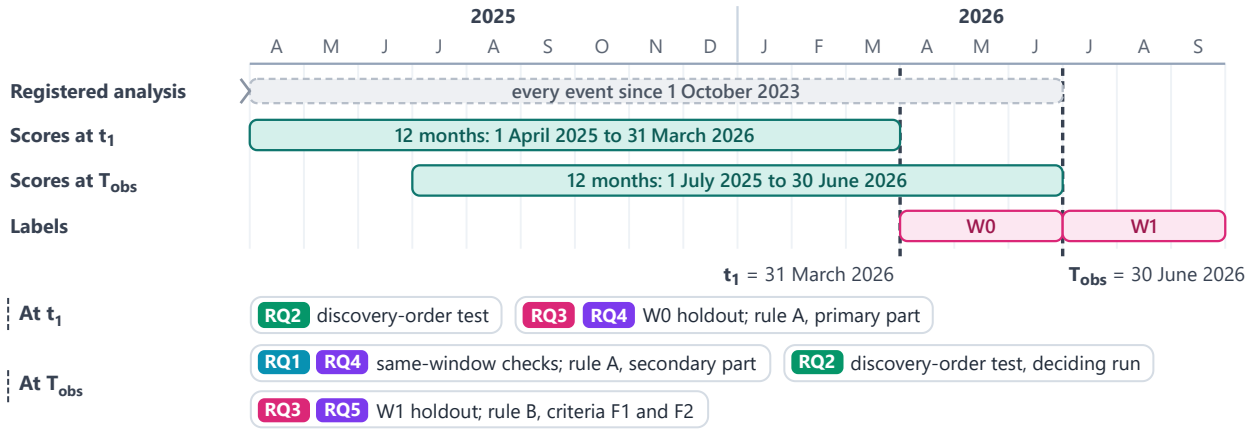



Figure 1.2: Freezes, score windows and label windows. The main analysis scores the twelve months before each freeze, $t_1$ (31 March 2026) and $T_{\text{obs}}$ (30 June 2026); W0 counts its labels from April to June 2026 and W1 from July to September 2026, and the registered analysis takes its scores from every event since 1 October 2023. Below the time line are the evaluations that read each freeze, with the research questions they answer: at $t_1$ the discovery-order test and the W0 holdout with the primary part of rule A; at $T_{\text{obs}}$ the same-window checks with the secondary part of rule A, the deciding run of the discovery-order test and the W1 holdout with criteria F1 and F2 of rule B.

The analysis plan and the registration of W1 define the registered analysis, with amounts in raw base units of every ERC-20 token, scores from every event since 1 October 2023 and graphs that hold every account of the cohort's events. After the registered results were known, the revised analysis was defined: amounts in US dollars on five tokens; scores from the twelve months before each

freeze; graphs whose nodes are contracts other than smart accounts, with edges read from the whole chain; each allowance counted up to its owner's balance at the freeze, with balances reconstructed from `Transfer` logs; constants set on the contract graph; and the discovery-order test. This revised analysis, the contract-graph analysis, is the main one (Section 3.5.3). Its verdicts under rule A and under the two criteria of rule B that the thesis applies are a re-analysis, not registered tests, and the registered analysis gave the same readings (Appendix 8.6).

## 1.5 Research objectives

The research has five objectives, one for each of RQ1–RQ5. Each is stated with its measurement and with the criterion that Chapter 5 applies when judging whether it was met.

O1. Establish whether EndorseRank and AWP order the contracts of the same cohort differently, how closely each score agrees with each construct in the same window, and what each walk costs, C-PR included (RQ1). Measurement: Kendall $\tau$ between the two score vectors ($n = 6{,}443$) and family-level Kendall $\tau$ for every score and construct, with 95% bootstrap intervals; solve time on the whole contract graph at $T_{\text{obs}}$ over five timed runs, memory, iterations, $|V|$ and $|E|$ under one protocol. Criteria: the interval of the agreement between the two walks contains neither $0$ nor $1$; a score agrees with a construct when its interval on that family excludes $0$, and agreement with the construct of its own edges is read as an intended-construct check; a time ratio is reported only with the ratio of $|E|$.

O2. Test whether a crawl ordered by C-PR on the contracts read so far reaches the contracts with the most user-account demand sooner than a breadth-first crawl and than a crawl ordered by the number of contract backlinks (RQ2). Measurement: crawls of the contract graph from 20 seed contracts under each ordering; for $I_T$ and for $I_A$, $\text{AUC}_{10\%}$, the normalized area under the curve of hot contracts reached over the first 10% of visits, the hot contracts being roughly the top 1% of the crawl's universe by that importance; two paired contrasts, $D_1$ (C-PR′ minus breadth-first) and $D_2$ (C-PR′ minus the backlink count $\text{IB}'_U$), with 95% bootstrap intervals.

Table 1.2: Summary of hypotheses, research questions, and empirical claims.

| Item | Addresses | Reported in |
|---|---|---|
| H1–H3 / RQ1 | The two single-layer walks: different orderings, agreement with the checks of each edge type, and the cost of each walk | §4.2, §4.3 and §4.4; Tables 4.5, 4.6, 4.3 and 4.4; Claims C5 and C6 |
| RQ2 | The discovery-order test: the crawl ordered by C-PR′ against a breadth-first crawl and a crawl ordered by contract backlinks | §4.6 and Appendix 8.4.4; Tables 4.12, 4.13 and 4.14; Figure 4.6; Claim C1 |
| RQ3 | Out-of-window prediction of new approval pairs and new transfer senders; each walk against the raw count of the edges it smooths | §4.7 and §4.8.4; Tables 4.15, 4.16, 4.17, 4.20, 4.18, 4.22 and 4.23; Claims C2 and C4 |
| RQ4 | The coupled walk against each of its layers walked alone, in window and out of window (rule A) | §4.8 and §4.8.2; Tables 4.5, 4.19 and 4.16; Claim C3 |
| RQ5 | The allowance layer added to a transfer walk on new data (two of rule B's registered criteria, F1 and F2) | §4.8.3; Tables 4.20 and 4.21; Claim C3 |
| C1 | In the deciding run the crawl ordered by C-PR′ reaches the contracts with the most user-account demand sooner than a breadth-first crawl but not sooner than a crawl ordered by contract backlinks, which is sooner: the answer to RQ2 is no. Outside the decision, in the other reveal mode, C-PR′ leads both comparators | Tables 4.12, 4.13 and 4.14 |
| C2 | Out of window, EndorseRank ranks new approval pairs below the in-approve degree; AWP ranks new senders slightly above the transfer in-degree on the whole cohorts but below it among the W0 spenders that had received a transfer from another contract; C-PR lies above the union in-degree on approval pairs, below it on senders, and below the degree of each label's own edge type | Tables 4.15, 4.16, 4.17, 4.18, 4.22 and 4.23 |
| C3 | Rule A, applied to the scores of the contract graph, is not met in either part; of rule B's two applied criteria F1 holds and F2 fails, so rule B is not met | Tables 4.16, 4.19, 4.21 and 8.16 |
| C4 | Beside the rules, C-PR ranks new approval pairs below EndorseRank and above AWP and S-PR, and new senders above EndorseRank and below AWP | Table 4.22; §4.8.4 |
| C5 | Same-window agreement of each single-layer walk with the checks of its own edge type, partly built in, and different orderings of the cohort by the two walks | Tables 4.5 and 4.6; Figure 4.4 |
| C6 | Runtime and graph size of each score: speed follows graph size and the iterations each walk needs, not the algorithm; C-PR takes slightly longer than AWP | Table 4.3 |

Criterion: the decision rule stated in the protocol of the test (Section 3.7), read at the freeze of 30 June 2026 in the primary reveal mode: the verdict on an importance is V1 when both intervals lie above $0$, V2 when only that of $D_1$ does and V3 otherwise, and the answer is the weaker of the two verdicts.

O3. Test whether scores frozen at a freeze predict the new approval pairs and new transfer senders

that spender contracts gain afterwards, and whether any walk beats the raw count of the edges it smooths (RQ3). Measurement: Kendall $\tau$ with 95% intervals between every score at the freeze, the raw counts included, and both labels, in W0 ($n = 7{,}864$ contract spenders) and W1 ($n = 7{,}915$), with paired contrasts of EndorseRank against the in-approve degree, of AWP against the transfer in-degree and of C-PR against the union in-degree and both single-layer degrees. Criterion: a score predicts a label when its interval excludes $0$, and a walk beats a count only when the interval of the paired difference lies above $0$.

O4. Test whether one walk across both edge types beats each of its layers walked alone (RQ4). Measurement: paired intervals on $\Delta\tau$ between C-PR at $\lambda = 0.5$ (with $0.25$ and $0.75$ as sensitivity values) and its two layers walked alone, and the same for its ablation S-PR. Criterion: rule A, fixed in the analysis plan before any log of the study was extracted: C-PR does no worse than its allowance layer on new approval pairs and better than its transfer layer on new senders in W0, and in the same window lies within the interval of the leading layer on the transfer and allowance families and above both layers on the stability family. The rule is applied to the scores of the contract graph as a re-analysis.

O5. Test whether adding the allowance layer to a walk over the transfers improves it on new data (RQ5). Measurement: paired contrasts of C-PR with its transfer layer walked alone in W1. Criterion: two of the criteria of rule B, registered after the W0 results were known and before the study extracted the W1 logs: under F1 C-PR beats the transfer layer on new approval pairs, and under F2 it is not worse by more than 0.01 on new senders. They are applied to the scores of the contract graph as a re-analysis, with the two registered sensitivity analyses (AWP in place of the transfer layer; contracts outside a graph kept as isolated nodes). The other contrasts of C-PR are reported beside both rules and decide neither.

## 1.6 Research questions

Three hypotheses go with the first research question. H1: on the same set of contracts, EndorseRank and AWP produce different orderings. H2: each score agrees with the checks built from the edges it walks on. H3: on the same contract graph, a score's computation time grows with the size of the layers it walks, so EndorseRank, on the sparser allowance layer, is the fastest to compute. H1–H3 hold largely by construction and act as checks of the instruments. The falsifiable tests are RQ2–RQ5. RQ4 and RQ5 follow the rules of the analysis plan and of the registration, applied to the scores of the contract graph as a re-analysis; RQ2 and RQ3 follow criteria defined after the registered results were known.

RQ1. Do EndorseRank and AWP, the walks over allowance edges alone and over transfer edges alone, order the same contracts differently, does each agree most with current checks of its own edge type, and what does each walk, the coupled one included, cost?

RQ2. Does a crawl ordered by C-PR on the contracts read so far reach the contracts with the most user-account demand sooner than a breadth-first crawl and than a crawl ordered by the number of contract backlinks? (the discovery-order test)

RQ3. Do scores frozen at a freeze predict later new approval pairs and new transfer senders, and does any walk, the coupled one included, beat the raw count of the edges it smooths?

RQ4. Does the coupled walk rank contracts better than either of its layers walked alone, in window and out of window? (rule A)

RQ5. Does adding the allowance layer improve a transfer walk on new data, gaining on later approvals without losing on later inflow? (two of rule B's registered criteria, F1 and F2)

RQ2 and RQ3 measure on the chain, by the demand of user accounts and by later labels, because the study had no label of a contract's importance from outside it. Rule B, the rule for RQ5, came after the W0 labels had been seen, so it is judged only on W1 (Section 3.8.1).

## 1.7 Summary of the literature review

Chapter 2 reviews four areas. In graph ranking, PageRank (Page et al., 1999) ranks nodes by recursively propagated endorsement, EigenTrust (Kamvar et al., 2003) and TrustRank (Gyöngyi et al., 2004) adapt it to peer-to-peer trust and web spam, and crawl-ordering experiments ask how soon an ordering leads a crawler to the important pages (Cho et al., 1998). In on-chain reputation, H.-D. Do and Do (2023) apply PageRank and HodgeRank to Ethereum transactions, and AWP (T. Do et al., 2023) weights transaction edges by value and age and checks its ranking against inbound transfer count and value, statistics drawn from the same data as its graph. D. Lin et al. (2025) propagate account risk through the transaction network, and surveys of blockchain trust and reputation systems (Bellini et al., 2020) and of reputation systems built on smart contracts (Almasoud et al., 2020) describe designs that rest on transfers or explicit ratings. Deployed DeFi tooling such as Human Passport instead aggregates on-chain and off-chain credentials to decide whether an address belongs to a distinct person (Human Passport, n.d.). None of this work ranks contracts on the relations among contracts alone or takes the ERC-20 allowance (Vogelsteller & Buterin, 2015) as a ranking edge; address clustering uses approvals only to link addresses of one entity (Victor, 2020). In DeFi risk and measurement, studies of wash trading (Victor & Weintraud, 2021), extractable value (Qin et al., 2022), scam tokens (Xia et al., 2021) and airdrop farming (Luo et al., 2025) show that much transfer activity is artificial. In measurement theory, construct validity (Cronbach & Meehl, 1955) separates what a score measures from external criteria, and the bootstrap (Efron, 1979) gives distribution-free intervals for rank statistics. The gap (Section 2.22) is a ranking that reads the relations among contracts with a walk over allowance and transfer edges, and a test of whether it finds the contracts that users rely on, beside the raw counts, with confidence intervals, paired differences, a fixed timing protocol and temporal holdouts.

## 1.8 Summary of the research methodology

Chapter 3 applies a quantitative, comparative design to public ledger data. ERC-20 `Approval` and `Transfer` logs on Arbitrum One from 1 October 2023 to 30 September 2026 are extracted from Google BigQuery in two scans, the second after W1 was registered. The main analysis takes from them every event of five tokens from 1 April 2025, types each address by its transactions and by its code at one pinned block, and keeps the events between two contracts in the twelve months before each freeze, in US dollars, with each allowance counted up to its owner's balance; these definitions were made after the registered results were known (Section 3.5.3). One power-iteration PageRank implementation computes every walk. The discovery-order test crawls the contract graph of each freeze under each ordering and compares how soon the crawls reach the contracts with the most user-account demand. Each score is also compared with Spearman's $\rho$ and Kendall's $\tau$ against same-window transfer, allowance and stability proxies and, out of window, against new approval pairs and new transfer senders on the spender contracts of W0 and W1, with the raw counts at each freeze as baselines. Except in the value-slope and sample-definition checks, every coefficient has a 95% interval from 400 paired resamples (seed 42), and method comparisons are paired. The code, configuration and machine-readable summaries behind every table are in a public repository (Section 3.14), and the study uses only public data (Section 3.17).

## 1.9 Contributions

The thesis makes five contributions to the study of on-chain reputation. EndorseRank and AWP both apply weighted PageRank (Page et al., 1999) to a single edge type, each with its own restart distribution; the coupled walk C-PR changes the transition rule itself. Most results are negative for the walk and are stated with each contribution.

- Coupled walk over allowances and transfers (C-PR): one random walk over a US-dollar allowance layer, with each allowance counted up to its owner's balance and capped, and a US-dollar transfer layer, which at each contract mixes the two transition rows with a fixed

weight $\lambda$ over the layers that contract occupies; S-PR is its seeded ablation. No reputation method reviewed couples the two relations (Section 2.22).

- A discovery-order test for a ledger ranking: the crawl-ordering experiment of Cho et al. (1998) repeated on a graph of contracts, with importance measured on the events of user accounts, which no ordering reads, and a decision rule stated in its protocol. The answer to RQ2 is no. The verdict is V2 on both importances: a crawl ordered by C-PR′ reaches the contracts with the most user-account demand sooner than a breadth-first crawl but not sooner than a crawl ordered by the number of contract backlinks, and the backlink ordering was sooner (Claim C1). Outside the decision, the allowance layer adds a small, resolved amount to C-PR′, and in the reveal mode that shows out-edges only, which carries no verdict, C-PR′ is sooner than both comparators.
- Its test under the rules of the plan and the registration: rule A, fixed with $\lambda$ in the analysis plan, and the two applied criteria of rule B set C-PR against its own layers walked alone; only these contrasts change the coupling and nothing else. Applied to the scores of the contract graph, rule A is not met in either part. Rule B is not met. The thesis applies two of its registered criteria, F1 and F2; under the registration the rule holds only if each of its criteria holds. F1, the gain over the transfer layer on new approval pairs in W1, holds, and F2 fails because the loss on new senders exceeds the margin (Claim C3). Beside the rules, C-PR ranks later approval pairs below EndorseRank and above AWP and its ablation S-PR, and later senders above EndorseRank and below AWP (Claim C4).
- The balance-capped allowance between contracts as a ranking edge (EndorseRank), the allowance reference: an edge $u \to v$ stands for the latest in-window ERC-20 allowance that the spender contract $v$ holds from the owner contract $u$, valued in US dollars up to the owner's balance and weighted as AWP weights a transfer. The edge records a delegation of spending rights that the owner can back, not a history of transfers, and no reputation method reviewed ranks on it (Section 2.22). Between contracts the layer is thin: most latest allowances are capped to zero, and at $T_{\text{obs}}$ one owner contract holds most of the pairs that remain (Tables 8.5

and 4.1). At depth one the walk has a closed form, $c\,(1 + d\,\delta(v))$ with $\delta(v)$ the sum of the transition probabilities into the spender $v$, $c$ the score common to nodes without in-edges and $d$ the damping factor; it holds to within 1% for every depth-one spender of the cohort but for only a minority of the deeper ones (Section 5.3.2).

- Evaluation of every walk against the count it smooths: every score is checked against the same proxies on one cohort under the protocol used with AWP (T. Do et al., 2023), against the approval pairs and senders that arrive after a freeze, and in discovery order, each time beside the raw counts, with 95% bootstrap intervals and paired contrasts (Efron, 1979). On these data a count wins more often than a walk: the count of contract backlinks reaches the important contracts sooner than C-PR′, EndorseRank ranks later approval pairs below the in-approve degree with either restart rule, and C-PR ranks later approval pairs above the union in-degree but later senders below it. AWP is the exception: its crawl is sooner than the crawl ordered by transfer backlinks, and it ranks later senders slightly above the transfer in-degree on the whole cohorts, though below it among the W0 spenders that had received a transfer from another contract (Claims C1 and C2).

## 1.10 Scope and limitations

The study extracts Arbitrum One from 2023-10-01 to 2026-06-30, W1 adds July to September 2026, and the main analysis scores the twelve months before each freeze, on the matched cohort and on the contract spenders of each freeze. The graphs hold the five-token events between contracts of the whole chain, and no user account moves a score. Account type is read from an address's transactions and from its code at one pinned block, not at the time of each event, and the rule for smart accounts recognizes Safes and ERC-4337 accounts, so kinds that it misses stay in the graphs as contracts (Appendix 8.4.3). The allowance layer between contracts is thin, and at $T_{\text{obs}}$ one owner contract holds most of its pairs (Table 4.1). About a third of the matched cohort and more than a quarter of each spender cohort end no pair between contracts and score zero under every score and

baseline (Table 4.2). The verdict of the discovery-order test is read in the reveal mode that the rule of its protocol made primary; in the other mode the contrast with the backlink count has the opposite sign (Table 4.12). No claim is made that the results generalize beyond Arbitrum One, the five tokens or the months studied. The amounts in US dollars on five tokens, the twelve-month score windows, the contract graph with its balance cap and the discovery-order test were defined after the registered results were known (Section 3.5.3).

## 1.11 Research roadmap

Chapter 2 reviews graph-based on-chain reputation, crawl ordering, ERC-20 allowance semantics and the reputation tooling deployed in DeFi, and develops the theoretical framework. Chapter 3 defines the account types, the tokens and prices, the contract graph and its balance cap, the scores, the discovery-order test (Section 3.7), the temporal holdouts, the construct checks, the timing protocol, and the standards for reproducibility and ethics. Chapter 4 reports the data of the contract graph, the benchmark (Claim C6), the same-window alignment of every score and the divergence of the two single-layer walks (C5), robustness checks, the discovery order (Section 4.6; C1), the temporal holdouts of the single-layer walks (C2), and the coupled walk under its rules and beside every other score (C3 and C4); Appendix 8.4.3 describes the account types and balances, Appendix 8.4.4 gives the further tables of the discovery-order test, Appendix 8.5 sets EndorseRank and AWP side by side, and Appendix 8.6 reports the registered analysis. Chapter 5 interprets the findings by research question and objective, works out their implications and weighs threats to validity and limitations. Chapter 6 answers RQ1–RQ5, states the contribution to knowledge and sets out future work. Figure 1.3 draws this plan.

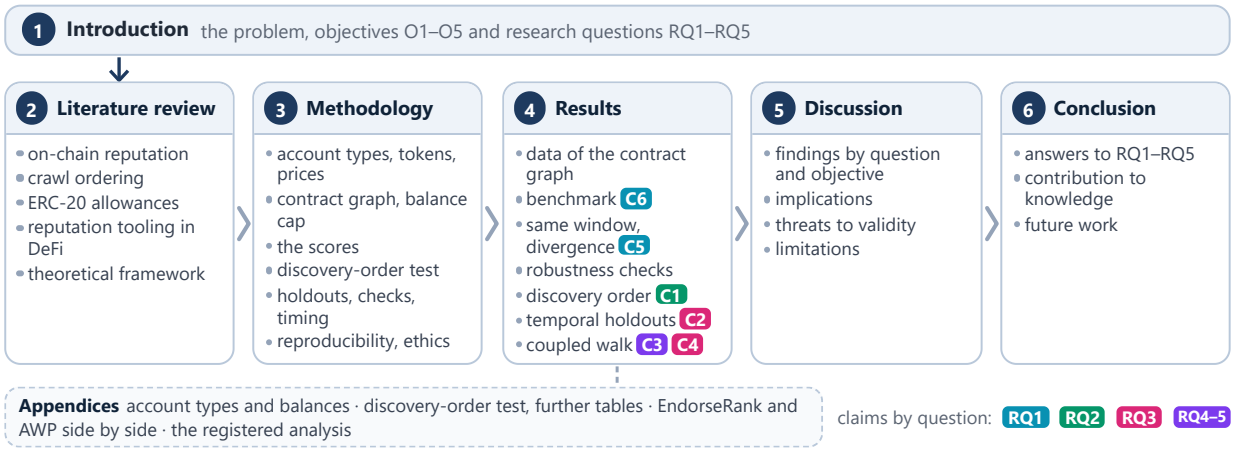



Figure 1.3: Research roadmap. Chapter 1 states the problem, the objectives and the research questions, and Chapters 2 to 6 follow from left to right with what each holds; the four appendices that the roadmap names are drawn below Chapter 4. Each claim of Chapter 4 is coloured by the research question it answers (Table 1.2).